

BOGD, Mongolia

WMO: 443380

Lat: 44.6742N

Lon: 102.1756E

Elev: 7156

StdP: 11.27

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-16.8	-28.7	1.4	-16.1	-24.2	1.9	-13.0	28.8	20.7	25.9	20.1	0.9	160	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	88.3	61.6	85.6	59.9	82.9	58.6	65.6	81.2	63.4	79.4	61.6	77.5	10.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.5	103.1	73.3	57.9	94.0	70.7	55.8	86.9	69.3	34.8	81.7	32.9	79.8	31.2	77.6	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
27.1	22.9	20.8		-27.0	93.7	5.7	3.3	-31.1	96.0	-34.4	98.0	-37.6	99.8	-41.8	102.2	
				-27.3	66.5	5.6	4.4	-31.3	69.7	-34.5	72.3	-37.6	74.7	-41.7	78.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	2.6	10.4	27.3	44.6	55.8	65.7	70.9	67.4	56.3	40.7	22.8	9.1
	DBStd	25.20	11.08	11.71	11.90	9.96	9.04	7.07	5.70	6.07	8.21	9.42	10.21	8.84
	HDD50	5977	1468	1109	704	216	48	1	0	0	36	310	817	1268
	HDD65	9706	1933	1529	1168	613	304	74	12	43	277	754	1266	1733
	CDD50	2186	0	0	1	53	228	471	648	541	223	21	0	0
	CDD65	443	0	0	0	1	19	94	195	119	15	0	0	0
	CDH74	4480	0	0	0	6	201	1031	1984	1147	111	0	0	0
	CDH80	1301	0	0	0	0	31	291	663	308	8	0	0	0
Wind	WSAvg	7.9	5.6	6.8	8.4	9.9	10.8	8.0	7.7	7.2	8.0	7.9	7.7	6.3
Precipitation	PrecAvg	5.9	0.1	0.1	0.2	0.4	0.5	1.0	1.2	1.1	0.7	0.3	0.2	0.2
	PrecMax	11.3	0.4	0.3	0.9	1.4	1.9	3.1	3.2	3.5	3.1	0.9	0.7	0.7
	PrecMin	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.2	0.1	0.1	0.2	0.3	0.4	0.7	0.8	1.0	0.9	0.2	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	41.2	58.8	73.5	83.5	90.7	94.4	90.5	80.9	67.6	51.9	36.1
		MCWB	25.1	30.8	39.6	50.7	52.2	62.9	66.8	61.9	55.3	47.9	38.1	27.2
	2%	DB	29.2	36.8	54.1	69.3	79.4	86.6	89.1	86.4	77.1	64.0	47.2	31.7
		MCWB	21.4	27.9	37.8	46.6	52.4	60.6	62.5	60.3	54.1	46.5	35.1	23.9
	5%	DB	24.5	32.9	49.9	65.6	75.5	83.2	86.4	83.5	74.1	59.8	42.8	27.5
		MCWB	18.1	24.9	35.2	45.3	50.8	58.0	60.7	59.0	52.1	43.3	31.9	21.3
	10%	DB	20.2	29.0	45.6	61.7	71.8	80.0	83.7	80.4	70.5	56.1	39.0	23.7
		MCWB	15.3	22.4	32.9	43.0	49.3	57.2	59.9	57.9	50.8	41.1	29.6	18.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	31.9	41.6	53.7	58.2	66.7	70.9	66.4	58.6	51.0	38.8	27.6
		MCDB	33.3	40.2	55.1	70.8	76.9	86.7	88.6	80.3	75.3	65.5	50.7	35.6
	2%	WB	21.6	28.2	38.2	49.2	54.7	63.8	66.8	64.2	56.4	47.5	35.5	24.0
		MCDB	28.9	36.4	53.0	65.9	73.1	81.4	82.9	78.7	71.1	61.1	46.6	30.7
	5%	WB	18.2	25.0	35.6	46.1	52.7	61.4	64.2	62.3	54.5	44.3	32.4	21.5
		MCDB	24.2	32.5	49.0	63.0	71.4	78.5	79.9	77.8	68.5	58.5	42.2	27.4
	10%	WB	15.2	22.4	33.0	43.6	50.7	59.4	62.1	60.5	52.5	41.8	29.8	18.6
		MCDB	19.9	28.8	45.2	60.3	69.2	75.9	78.0	75.5	67.6	54.8	38.5	23.6
Mean Daily Temperature Range	5% DB	MDBR	21.2	22.1	22.8	21.4	21.4	21.1	20.2	20.6	20.5	20.6	19.4	20.2
		MCDBR	23.9	23.8	24.7	24.4	24.6	24.7	24.0	24.0	22.5	22.6	21.2	22.0
		MCWBR	18.2	17.4	15.3	13.3	12.3	11.3	9.3	9.7	10.4	13.0	13.8	16.9
	5% WB	MCDBR	23.5	23.3	24.2	23.8	23.0	22.8	20.8	19.8	20.8	22.1	20.2	21.5
		MCWBR	18.2	17.4	15.8	14.2	12.9	12.2	10.3	8.9	10.8	13.8	13.7	16.6
Clear-Sky Solar Irradiance	taub	0.225	0.250	0.286	0.321	0.337	0.350	0.343	0.327	0.298	0.273	0.240	0.225	
	taud	2.430	2.374	2.303	2.257	2.261	2.301	2.362	2.391	2.435	2.444	2.452	2.423	
	Ebn at Noon	295	303	302	298	295	291	291	293	295	290	285	283	
	Edh at Noon	22	28	35	41	42	40	37	35	31	26	21	20	
All-Sky Solar Radiation	RadAvg	662	1014	1514	1949	2261	2253	2130	1943	1648	1199	770	567	
	RadStd	55	84	68	84	108	123	98	90	62	45	58	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-4.66	N/A	-2.25	-2.54	N/A	N/A	N/A

Nomenclature: See separate page